

Data gaps: -

Ozone sensor:

Science Pump Corporation (SPC) ECC-5A/6A (Entire record)

Radiosonde:

VAISALA RS80/RS90/RS41 (Entire record)

Sensing Solution Type (SST):

1% KI, 1.0x (full) buffer (Entire record)

Algorithm Description:

Ozone is calculated as a partial pressure. PTU data from the sonde is not used directly in the calculation except in the pump correction.

$$PPOZ(nb) = 0.004307 * i * Temperature * t * pcf$$

where:

the constant is half the ratio of ideal gas constant to Faraday's constant.

i is the current from the sensor - background in uA.

t is the time in seconds to pump 100 CCs of air through the pump.

Temperature is the pump temperature (K).

Pcf is the pump correction factor to account for loss in pump efficiency at lower pressures.

Pre-flight procedures comply with the VAISALA handbook:

Expected Precision/Accuracy of Instrument:

Ozonesonde:

Accuracy	Precision	Resolution
+/- 5%	+/- 4%	~150m

Instrument History:

ECC Ozonesondes changes:

19920322 - 19970920 SPC ECC-5A

19970920 - current SPC ECC-6A

Radiosonde changes:

19920322 - 20021231: RS80

20030106 - 20061005: RS90 with two exceptions:

20051229 - 20060104: RS92

20061011 - 20180523: RS92

20180530 - current: RS41

Changes in the location of the pump thermistor are related to the 5A or 6A ozonesonde type:

5A: taped near to the pump but not at the pump

6A: positioned in the dedicated thermistor hole of the sonde